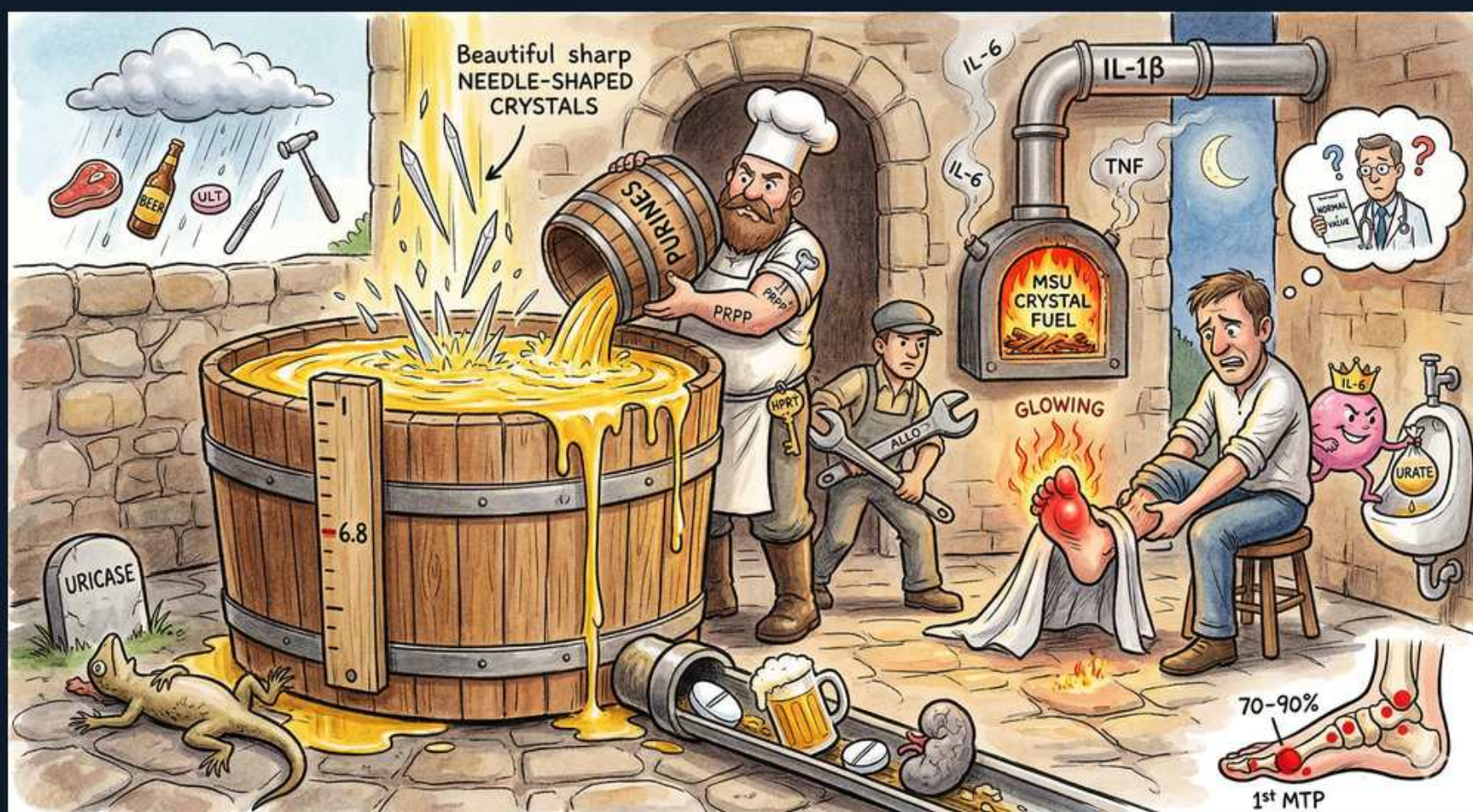


Gout/CPPD — Pathogenesis, Hyperuricemia, Flare



1. Needle-shaped crystals

WHAT YOU SEE

Yellow sharp needle crystals shooting from the barrel

WHAT IT ENCODES

MSU (monosodium urate) crystals are needle-shaped and **NEGATIVELY** birefringent under polarized light. They appear **YELLOW** when aligned **PARALLEL** to the slow axis of the red compensator and blue when perpendicular. Synovial aspirate is the gold-standard diagnostic test, showing intracellular crystals within neutrophils during a flare. Mnemonic: 'urAte = pARallel = yellow, Negative birefringence.'

BOARD TRAP

WRONG answer: calling rhomboid/positively birefringent crystals 'gout.' Those are CPPD (calcium pyrophosphate) = pseudogout, which is positively birefringent and **BLUE** when parallel to the compensator. Discriminator: needle + negative + yellow-when-parallel = gout; rhomboid + positive + blue-when-parallel = CPPD (and CPPD favors knees/wrists, chondrocalcinosis on X-ray).

2. Purines barrel

WHAT YOU SEE

Chef pouring a keg labeled PURINES into the vat

WHAT IT ENCODES

Purine catabolism is the substrate stream feeding uric acid: hypoxanthine → xanthine → uric acid, both steps catalyzed by xanthine oxidase (the drug target of allopurinol/febuxostat). Endogenous sources are high cell turnover (psoriasis, hemolysis, tumor lysis); dietary sources are red/organ meats, seafood, beer, and fructose. Fructose is unique because its metabolism consumes ATP and generates AMP, accelerating purine degradation to urate.

BOARD TRAP

WRONG answer: assuming strict purine-restricted diet alone will control gout. Diet changes lower serum urate only ~1 mg/dL — inadequate for tophaceous or recurrent disease. Discriminator: diet is adjunctive; definitive control requires urate-lowering pharmacotherapy to target <6 mg/dL.

3. Urate vat / 6.8

WHAT YOU SEE

Wooden vat overflowing, marked 6.8

WHAT IT ENCODES

The urate saturation/solubility limit in plasma is ~6.8 mg/dL at 37°C; above this, MSU supersaturates and precipitates in cool peripheral/avascular tissue (toes, ears, joints). Hyperuricemia is therefore defined as >6.8 mg/dL. The **TREATMENT** target is <6 mg/dL (and <5 mg/dL for tophaceous disease) to dissolve existing crystals. Cooler distal joints precipitate first — explaining podagra.

BOARD TRAP

WRONG answer: treating asymptomatic hyperuricemia with urate-lowering therapy. Most people with elevated urate never develop gout, and trials show no benefit treating asymptomatic levels. Discriminator: pharmacologic ULT is indicated only with clinical gout (≥2 flares/yr, tophi, urate stones, CKD), not for a high lab value alone.

4. PRPP

WHAT YOU SEE

PRPP label on the chef's apron

WHAT IT ENCODES

PRPP (phosphoribosyl pyrophosphate) synthetase **OVERACTIVITY** drives purine **OVERPRODUCTION** and is an X-linked cause of early hyperuricemia. The salvage enzyme HGPRT normally recycles hypoxanthine/guanine; its deficiency wastes PRPP toward de novo synthesis. Complete HGPRT deficiency = Lesch-Nyhan syndrome (X-linked): hyperuricemia, gout, self-mutilation, dystonia, intellectual disability.

BOARD TRAP

WRONG answer: attributing the majority of gout to overproduction. Overproduction (PRPP excess, HGPRT defect, tumor lysis, psoriasis) accounts for only ~10% of cases. Discriminator: a 24-hr urine urate >800 mg/day on a regular diet suggests overproducer (and contraindicates uricosurics like probenecid); the other ~90% are underexcretors.

5. Uricase lizard / grave

WHAT YOU SEE

Lizard by a tombstone reading URICASE

WHAT IT ENCODES

Humans and higher primates **LACK** uricase (urate oxidase) — the gene was silenced in evolution — so we cannot convert relatively insoluble uric acid into highly soluble allantoin. This is why humans get gout but most mammals do not. Recombinant uricase (rasburicase, pegloticase) exploits this missing pathway therapeutically.

BOARD TRAP

WRONG answer: using rasburicase for chronic gout. Rasburicase is for tumor lysis syndrome prophylaxis/treatment of acute hyperuricemia, not chronic gout (it's pegloticase that treats refractory tophaceous gout). Discriminator: both are uricase, but **BOTH** are contraindicated in G6PD deficiency because the H₂O₂ byproduct triggers hemolysis/methemoglobinemia.

6. Under-excretion (kidney)

WHAT YOU SEE

Kidney/pipe near the urinal

WHAT IT ENCODES

~90% of hyperuricemia is renal **UNDER**-excretion, not overproduction. Urate is freely filtered then largely reabsorbed via URAT1; net excretion is only ~10%. Drugs that reduce excretion (low-dose aspirin, thiazide/loop diuretics, cyclosporine, pyrazinamide, ethambutol), plus CKD, lactic acidosis, and ketoacidosis compete at the tubule and raise urate. Mnemonic for urate-raising drugs: 'CAN'T LEAP' (Cyclosporine, Alcohol, Nicotinic acid, Thiazides, Lasix/loops, Ethambutol, Aspirin-low-dose, Pyrazinamide).

BOARD TRAP

WRONG answer: choosing high-dose aspirin to lower urate or assuming all aspirin raises it. Aspirin is biphasic — **LOW** dose causes retention (hyperuricemia), **HIGH** dose is uricosuric. Discriminator: the boards expect low-dose ASA → ↑urate; do not stop cardioprotective aspirin for gout.

7. MSU crystal fuel furnace

WHAT YOU SEE

Glowing furnace labeled MSU CRYSTAL FUEL

WHAT IT ENCODES

Deposited MSU crystals are phagocytosed by resident macrophages and act as a danger signal that assembles and activates the NLRP3 inflammasome → caspase-1 cleaves pro-IL-1 β into active IL-1 β . IL-1 β recruits neutrophils that ingest more crystals, perpetuating the acute flare. This crystal→NLRP3→caspase-1→IL-1 β axis is the molecular core of gout and the rationale for IL-1 blockade.

BOARD TRAP

WRONG answer: thinking a NORMAL serum urate during an acute attack rules out gout. Urate often drops during a flare (acute-phase uricosuria/cytokine effect), so a normal level does not exclude it. Discriminator: diagnose on synovial crystals, not on the urate value drawn during the attack — recheck urate 2 weeks after resolution.

8. IL-1 β pipe

WHAT YOU SEE

Big pipe labeled IL-1B feeding the furnace

WHAT IT ENCODES

IL-1 β is THE pivotal cytokine of the acute gout flare downstream of NLRP3/caspase-1. This is the basis of anti-IL-1 therapy — anakinra (IL-1 receptor antagonist) and canakinumab (anti-IL-1 β monoclonal) — reserved for flares when NSAIDs, colchicine, and steroids are all contraindicated (e.g., advanced CKD with diabetes/heart failure).

BOARD TRAP

WRONG answer: reaching for anti-IL-1 biologics as first-line flare therapy. They are last-line/refractory agents. Discriminator: standard acute flare ladder is NSAIDs OR colchicine OR glucocorticoids first; IL-1 inhibitors only when those fail or are contraindicated.

9. IL-6 / TNF

WHAT YOU SEE

Smaller pipes labeled IL-6 and TNF

WHAT IT ENCODES

IL-6 and TNF- α are amplifying cytokines released alongside IL-1 β , driving systemic inflammatory features of severe flares (fever, leukocytosis, elevated CRP/ESR). They explain why a polyarticular gout flare can mimic systemic infection or sepsis with marked acute-phase reactants.

BOARD TRAP

WRONG answer: diagnosing 'sepsis/septic arthritis' from fever + leukocytosis + a hot joint without aspiration. A florid gout flare produces the same systemic picture. Discriminator: arthrocentesis distinguishes them — gout shows negatively birefringent crystals with WBC typically 20k–50k, whereas septic arthritis often shows >50k–100k with positive Gram stain/culture; the two can also coexist.

10. Hot red MTP joint

WHAT YOU SEE

Glowing inflamed foot joint wrapped in cloth

WHAT IT ENCODES

Acute gout = abrupt monoarthritis with severe pain, erythema, warmth, and swelling peaking within 12–24 hours, often waking the patient at night and so tender even a bedsheet hurts. Untreated flares self-resolve over days to weeks. Over time, untreated disease progresses to chronic tophaceous gout with destructive 'rat-bite'/punched-out erosions with overhanging edges on X-ray.

BOARD TRAP

WRONG answer: skipping joint aspiration because the presentation 'looks like classic gout.' Septic arthritis is the can't-miss mimic and a surgical emergency. Discriminator: always aspirate an acutely hot, swollen joint to send cell count, Gram stain/culture, AND crystal analysis before committing to a gout diagnosis.

11. 1st MTP / podagra

WHAT YOU SEE

Foot diagram, red dot at base of big toe, '70-90%'

WHAT IT ENCODES

The first metatarsophalangeal joint (podagra) is the classic initial site because it is the coolest, most peripheral, weight-bearing joint where urate precipitates most readily. It is the inaugural joint in ~50% and is involved in up to 90% of patients over the disease course. Other early sites: midfoot, ankle, knee.

BOARD TRAP

WRONG answer: assuming gout is always monoarticular at the great toe. Long-standing or polyarticular gout (especially in elderly women on diuretics) can present in the hands/DIPs and mimic osteoarthritis or RA. Discriminator: tophi over Heberden nodes/olecranon and crystal-positive aspirate confirm gout despite an atypical, polyarticular distribution.

12. Alcohol / beer trigger

WHAT YOU SEE

Beer bottle and meat among the trigger icons

WHAT IT ENCODES

Flare triggers cause rapid urate fluctuation in either direction: alcohol (esp. beer), purine-rich feasts, dehydration, new diuretics, trauma, surgery, acute illness, and — importantly — INITIATION of urate-lowering therapy. Rapid lowering mobilizes crystals from tophi and can paradoxically precipitate a flare.

BOARD TRAP

WRONG answer: stopping urate-lowering therapy when a flare occurs during the first weeks of treatment. The flare is expected crystal mobilization, not drug failure. Discriminator: CONTINUE the ULT through the flare, treat the flare symptomatically, and co-prescribe colchicine/NSAID prophylaxis for 3–6 months when starting ULT.

13. Beer mugs (mob)

WHAT YOU SEE

Row of beer mugs at the base

WHAT IT ENCODES

Beer is the worst alcoholic trigger (high purine/guanosine content PLUS ethanol, which increases ATP turnover and lactate that blocks urate excretion), spirits are intermediate, and wine is least associated. Fructose-sweetened sodas independently raise urate. Protective dietary factors: coffee, low-fat dairy, vitamin C, and cherries.

BOARD TRAP

WRONG answer: telling a gout patient that switching from beer to wine fully resolves risk. All alcohol contributes via lactate-mediated underexcretion; wine is only relatively safer. Discriminator: the high-yield protective foods are coffee, low-fat dairy, and cherries — not 'any alcohol in moderation.'